

Arunabha Majumder

Mechatronics & Robotics Engineer | Robotic Hardware | Pneumatics | Sensor Integration | Prototyping

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PROFESSIONAL SUMMARY

Hands-on Mechatronics and Robotics Engineer with a B.Tech in Mechanical Engineering, an M.Tech in Mechatronics and a PhD in progress at Aalborg University. Designs, builds and tests robotic hardware end to end: CAD, 3D printing and machined parts, assembly, integration of sensors, actuators, pneumatics and control systems, and rapid iteration through physical testing. Took robotic systems from concept to working hardware, including a hybrid-actuated exoskeleton (Best Paper Award, IFToMM ISRM 2026) and in-house pneumatic actuators, and developed sensorless force estimation for a 5-bar parallel robot. Independent, pragmatic and solution-oriented, with a builder's mindset; fluent in English (C1).

CORE SKILLS

- **Robotics & Automation:** Robotic hardware, robotic mechanisms, parallel robots, actuator design, wearable robotics
- **Pneumatics:** Pneumatic actuators, valves, pressure sensors, pneumatic control
- **Sensor & Control Integration:** Motors, encoders, force/torque sensors, DAQ, embedded control, hardware-software co-design, control systems (PID, neural-network-based gain scheduling)
- **CAD & Design:** SolidWorks (CSWA certified), Fusion 360, 3D modelling, technical drawings, FEA, Design for Assembly (DFA)
- **Manufacturing & Prototyping:** 3D printing (Bambu Lab, Ultimaker Cura), machined parts, hands-on fabrication, assembly in lab and workshop
- **Testing & Iteration:** Rapid prototyping, custom test rigs, physical testing, troubleshooting, data analysis (Python, MATLAB)

PROFESSIONAL EXPERIENCE

Research Engineer – Robotics & Actuator Development (PhD Project: VIEXO) | 2023 – Present *Aalborg University, Aalborg, Denmark*

- Designed, built and tested robotic hardware end to end: SolidWorks CAD and FEA, 3D-printed and machined parts, assembly, motor/encoder/force-sensor and DAQ integration, embedded control and custom automated test rigs.
- Cut energy consumption of a shoulder exoskeleton by more than 25% by designing a hybrid actuation system that combines a parallel spring with a 6 Nm motor; awarded Best Paper, IFToMM ISRM 2026.
- Removed the need for an external force/torque sensor in a 5-bar parallel robot by deriving a sensorless force-estimation method from the actuator torque–deflection characteristic, reducing cost, mass and integration complexity.
- Deployed an Assist-as-Needed hybrid controller on functional prototype hardware, reducing user muscle effort by more than 15% in tests with human subjects.
- Validated analytical stiffness models against physical hardware to below 10% RMS error, iterating designs through rapid prototyping and physical testing.
- Ran a 12-participant hardware test campaign across 3 assistance modes and 2 load conditions, including instrumentation (EMG, camera-based motion capture) and the end-to-end data pipeline.
- Taught university students and supervised Bachelor's (B.Tech) students (2023 – 2025).

M.Tech Researcher – Pneumatic Actuator Systems (Master's Thesis) | 2020 – 2022 *CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India*

- Fabricated pneumatic artificial muscle actuators in-house from raw materials and built the pneumatic test setup with valves, pressure sensors and DAQ.

- Reduced position-tracking error to 0.3–0.78% by designing and implementing a neural-network-based gain-scheduled PID controller for a non-linear pneumatic actuator, outperforming classical PID under varying load.

Mechanical Engineering Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019
Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, fabrication and user testing of a robotic rehabilitation exoskeleton; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.

HANDS-ON BUILD PROJECTS

- **Go-Kart (2017):** Led end-to-end design, fabrication and build of a custom Go-Kart from the ground up.
- **Efficycle (2018):** Worked in a multidisciplinary team to design and build a human-electric hybrid vehicle for the Efficycle competition.

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022
AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019
Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

CERTIFICATIONS & AWARDS

- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Major Project Award (2019); TATA Technologies Innovation Award – INR 100,000 Grant (2019)

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026, MMS*, vol. 213, Springer.
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

TECHNICAL TOOLS

- **CAD & Manufacturing:** SolidWorks, Fusion 360, FEA, Bambu Lab and Ultimaker Cura (3D printing)
- **Software & Control:** Python, MATLAB/Simulink, LabVIEW (basic), Arduino
- **Instrumentation & Testing:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, Pneumatic Control, Custom Rig Fabrication

LANGUAGES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

REFERENCES

Available upon request.